ZIJIAN WANG

Phone: +86 199 5339 5502 **Email:** zijian.wang@connect.polyu.hk Male | Age 23 | IELTS 6.5

Electronic & Information Engineering | Vision-Language-Action Models | Robotic Intelligence

RESEARCH PROFILE

M.Sc. candidate focused on embodied AI and AI4S, with a sole-first-author blind face restoration paper and hands-on experience developing Vision-Language-Action models for robotic pipetting.

EDUCATION

The Hong Kong Polytechnic University Sep. 2025 - Present

M.Sc. in Electronic and Information Engineering

GPA: 3.23/4.30 | Core Coursework: Big Data Computing, Video Technology, Wireless Data Networks

Chongqing University of Technology Sep. 2021 - Jul. 2025

B.Eng. in Computer Science and Technology

GPA: 3.73/5.00 | Core Coursework: Advanced Programming, Data Structures, Fundamentals of Robot Control

RESEARCH EXPERIENCE

Topology-Preserving and Frequency-Disentangled Blind Face Restoration Sep. 2025 - May 2026

The Hong Kong Polytechnic University

- Designed the methodology, optimized the network architecture, prepared key figures, and led manuscript development under faculty supervision.
- Integrated Hilbert serialization with frequency disentanglement and tunable weighting to flexibly regulate high-frequency texture injection for robust blind face restoration.
- Sole first author of *Hift-Former: Unifying Hilbert Serialization and Frequency Priors for Robust Blind Face Restoration*, accepted by the Eurographics Symposium on Rendering (EGSR, CCF-B); delivered an oral conference presentation.

Scientific Embodied Intelligence for Robotic Pipetting May 2026 - Present

Center for AI for Science and Engineering (AI4SE), Shenzhen Loop Area Institute (SLAI) | Supervisor: Prof. Deng-Ping Fan

- Fine-tune, train, and deploy Vision-Language-Action models; own data preprocessing, training configuration, inference deployment, experimental analysis, and FR3 robotic arm-electronic pipette integration tests.
- Resolved repeated movements from long waits and limited action horizons by adding task-phase annotations and redesigning model I/O and device-state representations, substantially improving long-horizon multi-actuator coordination.
- Developed a closed-loop tip-attachment framework using OpenCV localization, visual servoing, force feedback, and device-state detection; enabled autonomous rack localization, coordinate transformation, and stable attachment across varying rack positions while reducing fixed-scene and manual-calibration dependence.

HONORS & AWARDS

RoboCup and Robot Competition - First Prize (**Champion**) 2022

RoboCup and Robot Competition - Third Prize 2023

Freshman Scholarship, Chongqing University of Technology - Top **5%** 2021

SKILLS

English Proficiency & Academic Communication: IELTS 6.5; all undergraduate and postgraduate coursework, assignments, and examinations completed in English; academic literature review, scholarly writing, oral communication, research presentations, and independent study.

Model Training & Computing Platforms: SenseCore AI Computing Platform; independent environment configuration, data preprocessing, VLA fine-tuning, inference evaluation, and experimental analysis; multiple training cycles covering deployment, hyperparameter tuning, stability monitoring, troubleshooting, and reproducibility.

Robotic Systems & Dataset Construction: Franka Research 3, libfranka, ROS 2, force-feedback control, OpenCV-based visual localization, and visual servoing; robotic system debugging, trajectory execution, task-oriented data collection and annotation, system integration, and real-world experimental validation

王梓鉴

电话：+86 199 5339 5502　　**邮箱**：zijian.wang@connect.polyu.hk　　男 | 23岁 | 雅思 6.5

电子与信息工程 | 视觉语言动作模型 | 机器人智能

研究简介

电子信息工程硕士研究生，研究方向聚焦具身智能与人工智能赋能科学研究。以独立第一作者身份完成盲人脸修复论文，并具有视觉语言动作模型在机器人移液任务中的研发与部署经验。

教育背景

香港理工大学　　2025.09 - 至今

电子信息工程 硕士

GPA：3.23/4.30 | **核心课程**：大数据计算、视频技术、无线数据网络

重庆理工大学　　2021.09 - 2025.07

计算机科学与技术 学士

GPA：3.73/5.00 | **核心课程**：高级程序设计、数据结构、机器人控制技术基础

科研经历

盲人脸修复的拓扑保真与频率解耦研究　　2025.09 - 2026.05

香港理工大学

- 在导师指导下完成方法设计、网络架构优化、论文关键图表设计及论文撰写。
- 首次将希尔伯特序列化与频率解耦引入盲人脸修复，并融合可调权重，灵活控制高频纹理注入。
- 以独立第一作者身份完成论文《Hift-Former: Unifying Hilbert Serialization and Frequency Priors for Robust Blind Face Restoration》，获 Eurographics Symposium on Rendering（EGSR，CCF-B）接收并受邀口头汇报。

科学具身移液项目　　2026.05 - 至今

深圳河套学院科学与工程智能中心 | 指导教师：范登平教授

- 负责视觉语言动作（VLA）模型的微调、训练与推理，完成数据预处理、训练配置、推理部署和结果分析；参与 FR3 机械臂与电子移液枪系统联调及真实环境实验验证。
- 针对静止等待时间较长、模型预测时域有限引发的重复动作，以及机械臂与电子移液枪控制不稳定等问题，引入任务阶段标注，重新设计模型输入输出维度与设备状态表示，提升长时序任务中的多执行器协同能力。
- 针对人工示教难以稳定完成枪头插取的问题，提出融合 OpenCV 视觉定位、视觉伺服、力反馈与设备状态检测的闭环插取方案，实现不同位置枪头盒的自主识别、坐标转换与稳定插取，降低对固定场景和人工标定的依赖。

荣誉奖项

中国机器人大赛暨 ROBOCUP 机器人世界杯中国赛一等奖（**全国冠军**）　　2022

中国机器人大赛暨 ROBOCUP 机器人世界杯中国赛三等奖　　2023

重庆理工大学新生奖学金（前 5%）　　2021

技能特长

英文学术沟通能力：雅思 6.5；本科及硕士阶段课程、作业与考试均在全英文环境下完成；具备英文文献阅读、学术写作、口头表达、科研交流、学术汇报及独立学习能力。

模型训练与计算平台应用能力：熟练使用 SenseCore 商汤大装置计算平台，可独立完成环境配置、数据预处理、VLA 模型微调、推理评测与实验分析；具有训练任务部署、参数调整、稳定性监测、异常排查及实验复现经验。

机器人系统与数据构建能力：熟悉 FR3 机械臂、libfranka、ROS 2、力反馈控制、OpenCV 视觉定位与视觉伺服；具备机器人系统调试、轨迹执行、任务数据采集与标注、系统联调及真实环境实验验证能力。